

that demand for those parts is high, and thus, they're easily available everywhere. However, if your phone suffered failure of some weird chip or resistor that is uncommon, you're looking at a wait time to be able to get it, if at all. It will also be far more expensive to fix a rare problem than an obvious one like a cracked screen glass. But do the manufacturers want to provide this to you, or other services such as repair shops? No they don't and we will find out later why...

MINIATURISATION

Moore's Law might not really be a "law", but it seems to rule over everything in the technology world like a benevolent dictator. In its loosest interpretation, Moore's Law suggests that technology will drive things to be both smaller as well as better performing. This unstoppable march of progress in technology has branched off into so much more than Gordon Moore had ever envisioned, because he was talking about CPUs. Today, technology is all pervasive, and works hard to improve every aspect of our life, and those who have the brain-child to use technology in a novel way often go on to build fortunes just from an idea! All of this is made possible by the ever shrinking footprint technology demands.

The miniaturisation of technology allows us to place sensors in everything, to connect wirelessly, to have batteries last longer and to, obviously, make gadgets smaller and more portable. The flip side to this is that miniaturisation means a lot of integrated circuits (ICs) are tiny, and delicate, and border on being unrepairable except by the most skilled of hands. When you need the precision of robotics to build something terribly small, the average human hand loses the ability to improve on it in any way.

While it's allowing us to fit more computing power into smaller packages and using less and less raw material in the process, it's also killing off the ability to repair anything. Instead, even a tiny fault in an IC can result in having to purchase an entirely new IC, instead of just replacing the faulty component itself. More on this later. **d**

SAFETY AND SECURITY

WHAT THE BRANDS HAVE TO SAY ABOUT THE DESIRE TO MAKE RIGHT TO REPAIR INTO A LAW

Before we dive down the rabbit hole and start accusing brands of anything. Let's first take a look at what they have told legislators and journalists across the globe. It's unfair to assume that brands are only against third party repairing because of profits and good share prices. So what are the arguments that are put forth by brands?

SAFETY

This one is a no-brainer. Remember Samsung's Note 7 disaster? The one where batteries were randomly catching fire or exploding? It got so bad that you weren't even allowed on flights with the Note 7 and Samsung just recalled them all, costing them a ton of money – some estimates put that loss at \$17 billion! This includes the actual losses faced and the damage to the brand image. They have made sure to not repeat those mistakes again though, so maybe this is all in the past for them, except that we keep bringing it up as a reminder, much to their chagrin.

The point we're making is that no one wants their products exploding or catching fire and they certainly do not want anyone getting injured in the process, or else it spells curtains for the brand. One way to do this is to lock things down so that there is little possibility of a user tampering

with a device and putting themselves in harm's way. This is why we saw cellphones go from these devices anyone with a decent screwdriver kit could open up and look inside to just 10 years later, devices that need special knowledge and tools and experience to be able to open up. Our regular TearDown section (Page 44 this month) collaborates with iFixit to show you the innards of popular smartphones, and you can see just how complex and compact the innards are.

Another way to do this is to tightly control what devices work with a product, and using certification to, say, grant access to functions on a device. For example, Apple's MFi (Made For iPhone/iPad/iPod/iOS/iMac/<iAnything>) program certifies third party made accessories (including chargers and cables) to Apple's standards and in some cases allows them to include a chip that identifies them as such, so that Apple devices charge at full speed, or even charge at all. Apple claims to do this because they want their devices to last longer and be safe, and only MFi certified devices meet their approval. You might even void your warranty if you use a non-MFi charging accessory with your iOS device!

Now, certification is obviously a must, as it prevents fly-by-night manufacturers popping up and